

PRASENJIT KARMAKAR

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PROFESSIONAL PROFILE

I am a PhD Researcher at IIT Kharagpur and a Prime Minister's Research Fellow, with hands-on experience in turning research into real systems. My work spans embedded hardware, IoT sensor networks, and multimodal AI workflows deployed in real-world environments. Published at NeurIPS, ACM CHI, and IEEE PerCom on sensing and human-computer interaction.

TECHNICAL SKILLS

- **Programming Languages and Concepts:** C/C++, Java, Python, Object Oriented Programming
- **Artificial Intelligence:** Data Analytics, Machine Learning, Deep Learning, Reinforcement Learning, Quantization-aware Training and Deployment, MLOps, Model Drift Management
- **NLP and Generative AI:** LLM and VLM Systems, Generative AI, Fine-tuning, Embodied AI, LLM Prompt-tuning, Multimodal Sensor and Language Fusion, KV-Caching, Inference Optimization
- **Libraries and Frameworks:** Scikit-Learn, Numpy, Pandas, Matplotlib, Scipy, Seaborn, PyTorch, TensorFlow, TFLite, Slurm, Unsloth, Hugging Face Transformer, PEFT, Transformer Reinforcement Learning, RLHF, DPO
- **Applied Systems:** IoT and Sensor, AR/VR, Wearables, Interaction Design, User Studies, Distributed Systems, Embedded Systems, Prototyping, 3D Design, Blender, Unity3D, KiCAD
- **Research and Communication:** Paper and Patent Drafting, Proposal Writing, Teaching

WORK EXPERIENCE

- **Singapore Management University | Visiting Researcher** **Singapore**
Collaborated in Embodied AI and LLM task planning projects *July 2025 – Jan 2026*
Skills: Multimodal AI, LLM/VLM, GenAI, Finetuning, Embodied AI
- **Tata Consultancy Services | Systems Engineer** **Bangalore, India**
Contributed as a Frontend Web Developer in the BFSI Sector *Nov 2020 – Apr 2022*
Skills: Lit-Element, React, Flask, Spring Boot
- **National Institute of Technology | Research Intern** **Durgapur, India**
Contributed as an ML Researcher to Develop Smart-city Applications *Sep 2019 – Sep 2020*
Skills: Embedded System Programming, Python, Data Science, MLOps
Outcome: Intelligent Bus Transport [[BuStop](#)], City-scale Air Quality Monitor [[AQaMoHo](#)]

EDUCATION

- **Indian Institute of Technology Kharagpur** **Kharagpur, India**
Pursuing Ph.D - Department of Computer Science and Engineering; CGPA: 8.95 *Jul 2022 – Dec 2026*
Prime Minister's Research Fellow (since April 2023)
Broad Research Area: Ubiquitous Computing for Environment Sensing
Ph.D Advisor: *Dr. Sandip Chakraborty* [[UbiNet](#)][[Lab Intro](#)]
- **Maulana Abul Kalam Azad University of Technology** **Kolkata, India**
B.Tech - Department of Computer Science and Engineering; CGPA: 9.13 *Jul 2016 – Jul 2020*
Thesis: Reinforcement Learning in Continuous Multi-joint Environments [[Report](#)]

FILED PATENTS

- I1 Prasenjit Karmakar, Sandip Chakraborty.** *An End-to-End System for Multi-modal Physiological and Environmental Exposure Monitoring.* India Patent Application No: 202631101632. Filed on: Aug 22, 2026.

- I2 Prasenjit Karmakar**, Sai Deepak Reddy Mara, Mupparaju Venkata Sai Varun, Sandip Chakraborty. *An End-to-End System for Modulating Ambient Lights with Environmental Sensors and LLM-guided Co-learning*. India Patent Application No: 202631099342. Filed on: Aug 17, 2026.
- I3 Prasenjit Karmakar**, Sandip Chakraborty. *A System with Framework Architecture for Distributed Air Quality Monitoring in Indoor Spaces*. India Patent Application No: 202531006180. Filed on: Jan 24, 2025. (Published)
- I4 Prasenjit Karmakar**, Sandip Chakraborty. *A Framework for Monitoring and Ventilating Personal CO₂ Bubbles through a Wrist Wearable and Virtual Reality Application*. India Patent Application No: 202531001296. Filed on: Jan 06, 2025. (Published)

RESEARCH PUBLICATIONS

- J1 Prasenjit Karmakar**, Swadhin Pradhan, Sandip Chakraborty. *Exploring Indoor Air Quality Dynamics in Developing Nations: A Perspective from India*. ACM Journal on Computing and Sustainable Societies 2024 [[Paper](#)][[Demo Video](#)][[Project Page](#)][[Google's SCAI Award](#)]
- J2** Argha Sen, Avijit Mandal, **Prasenjit Karmakar**, Anirban Das, Sandip Chakraborty. *Passive Monitoring of Dangerous Driving Behaviors Using mmWave Radar*. Pervasive and Mobile Computing 2024 [[Paper](#)]
- J3** Prithviraj Pramanik, **Prasenjit Karmakar**, Praveen Kumar Sharma, Soumyajit Chatterjee, Subrata Nandi, Sandip Chakraborty, Mousumi Saha, Sujoy Saha. *AquaMoHo: Localized Low-Cost Outdoor Air Quality Sensing over a Thermo-Hygrometer*. ACM Transactions on Sensor Networks 2023 [[Paper](#)][[Codebase](#)]
- J4 Prasenjit Karmakar**, Vijay K Shah, Satyaki Roy, Krishnandu Hazra, Sujoy Saha, Subrata Nandi. *Reliable backhauling in aerial communication networks against UAV failures: A deep reinforcement learning approach*. IEEE Transactions on Network and Service Management 2022 [[Paper](#)]
- J5** Ratna Mandal, **Prasenjit Karmakar**, Soumyajit Chatterjee, Debaleen Das Spandan, Shouvit Pradhan, Sujoy Saha, Sandip Chakraborty, Subrata Nandi. *Exploiting Multi-modal Contextual Sensing for City-bus's Stay Location Characterization: Towards Sub-60 Seconds Accurate Arrival Time Prediction*. ACM Transactions on Internet of Things 2022 [[Paper](#)][[Codebase](#)][[Interactive Demo](#)][[Dataset](#)]
- C1 Prasenjit Karmakar**, Karthik Reddy, Sandip Chakraborty. *PoHAR: Understanding Hyperlocal Human Activities with Pollution Sensor Networks*. IEEE DCOSS-IoT 2026 [[Paper](#)][[Demo Video](#)][[Project Page](#)]
- C2 Prasenjit Karmakar**, Manjeet Yadav, Swayanshu Rout, Swadhin Pradhan, Sandip Chakraborty. *From Invisible to Actionable: Augmented Reality Interactions with Indoor CO₂*. ACM CHI 2026 (core-A*) [[Paper](#)][[Demo Video](#)][[Project Page](#)]
- C3 Prasenjit Karmakar**, Swadhin Pradhan, Sandip Chakraborty. *Indoor Air Quality Dataset with Activities of Daily Living in Low to Middle-income Communities*. NeurIPS 2024 D&B (core-A*) [[Paper](#)][[Dataset](#)]
- C4** Argha Sen, Avijit Mandal, **Prasenjit Karmakar**, Anirban Das, Sandip Chakraborty. *mmDrive: mmWave Sensing for Live Monitoring and On-Device Inference of Dangerous Driving*. IEEE PerCom 2023 (core-A*) [[Paper](#)][[Codebase](#)][[Demo Video](#)][[Project Page](#)]
- C5** Argha Sen, Anirban Das, **Prasenjit Karmakar**, Sandip Chakraborty. *mmAssist: Passive Monitoring of Driver's Attentiveness Using mmWave Sensors*. COMSNETS 2023 [[Paper](#)][[Codebase](#)][[Demo Video](#)]
- C6** Praveen Kumar Sharma, **Prasenjit Karmakar**, Soumyajit Chatterjee, Abhijit Roy, Santanu Mandal, Sandip Chakraborty, Subrata Nandi, Sujoy Saha. *Can I go for a roof walk today? know your housing's air quality from a thermo-hygrometer*. ACM BuildSys 2021 [[Paper](#)][[Demo Video](#)]
- P1 Prasenjit Karmakar**, Swadhin Pradhan, Sandip Chakraborty. *CookSense: What is Cooking in Your Kitchen?*. ACM Mobicom 2026 Work-in-Progress
- P2** Manjeet Yadav, **Prasenjit Karmakar**, Suchetana Chakraborty. *Understanding Behavioral Dark Patterns of High BMI Individuals*. UbiComp/ ISWC 2026 DiversityOne Workshop [[Paper](#)][[Demo Video](#)]

- P3** Neeraj Boddada, Sharvari Wanjari, Shashank Goud Boorgu, **Prasenjit Karmakar**, Sandip Chakraborty. *On-Device Emotion Recognition from Spoken Language in Embedded Devices*. IEEE PerCom 2025 Work-in-Progress [Paper][Demo Video]
- P4** Anuj Kakde, Yashwant Krishna Pagoti, Sarthak Nikumbh, **Prasenjit Karmakar**, Sandip Chakraborty. *Real-time Air Quality Monitoring and Context-aware Alert System*. COMSNETS 2025 Poster [Demo Video]
- P5** **Prasenjit Karmakar**, Swadhin Pradhan, Sandip Chakraborty. *Exploiting Air Quality Monitors to Perform Indoor Surveillance: Academic Setting*. ACM MobileHCI 2024 Late-Breaking-Work [Paper][Demo Video]
- P6** **Prasenjit Karmakar**, Ajay Kumar Meena, Kushal Natani, Sandip Chakraborty. *Multimodal Sensing for Predicting Real-time Biking Behavior based on Contextual Information*. IEEE PerCom 2024 Work-in-Progress [Paper][Codebase][Demo Video]
- P7** Ratna Mandal, **Prasenjit Karmakar**, Abhijit Roy, Arpan Saha, Soumyajit Chatterjee, Sandip Chakraborty, Sujoy Saha, Subrata Nandi. *Ad-hocBusPoI: Context Analysis of Ad-hoc Stay-locations from Intra-city Bus Mobility and Smartphone Crowdsensing*. ACM SIGSPATIAL 2020 Poster [Paper][Demo Video]

HONORS, AWARDS AND RECOGNITION

- | | |
|---|------------|
| • Awarded Prime Minister Research Fellowship | Mar 2023 |
| • Supported by Google's Award for Inclusion Research (USD 35K) | Oct 2023 |
| • Supported by Google's Award for Society Centered AI (USD 100K) | Sept 2025 |
| • Awarded LRN International Travel Grant | April 2026 |
| • Awarded ACM IARCS Travel Grant | April 2026 |
| • Awarded COMSNETS Travel Grant | Jan 2025 |
| • Awarded ACM COMPASS Travel Grant | Jul 2024 |
| • Volunteer at ICDCN, West Bengal, India | Jan 2023 |

OTHER PROFESSIONAL ACTIVITIES

- TA: Autumn 2026. Reinforcement Learning. **NPTEL** (IIT Madras) [Recorded Sessions]
- TA: Spring 2023, 2024, and 2026. Computer Networks and Internet Protocol. **NPTEL** (IIT Kharagpur)
- TA: Spring 2025. Introduction to Machine Learning. **NPTEL** (IIT Madras) [Recorded Sessions]
- TA: Fall 2024. The Joy of Computing using Python. **NPTEL** (IIT Ropar) [Recorded Sessions]
- TA: Autumn 2026. Machine Learning. IIT Kharagpur
- TA: Spring 2025, 2026. Computer Networks. IIT Kharagpur
- TA: Fall 2024. Ubiquitous Computing. IIT Kharagpur
- TA: Spring 2024. Distributed Systems. IIT Kharagpur
- TA: Spring 2023 and Fall 2023. Programming and Data Structure Lab. IIT Kharagpur
- Served as a reviewer in IEEE TMC, PMC, NeurIPS, ACM CHI, ACM CSCW, ACM JCSS, ACM MobileHCI

REFERENCE

- Prof. Sandip Chakraborty (**PhD Advisor**), IIT Kharagpur, India. sandipc@cse.iitkgp.ac.in
- Prof. Archan Misra. Singapore Management University, Singapore. archanm@smu.edu.sg
- Prof. Subrata Nandi. NIT Durgapur, India. subrata.nandi@cse.nitdgp.ac.in
- Dr. Swadhin Pradhan. Cisco Meraki, United States. swapradh@cisco.com